

VISION-BASED ROBOTIC BANANA SORTING SYSTEM FOR QUALITY CONTROL IN KETCHUP MANUFACTURING USING YOLOV11

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Abstract— Consistent raw material quality is a critical prerequisite for the reliable manufacturing of banana ketchup, particularly when utilizing the Saba banana variety. Conventional manual sorting workflows are labor-intensive, error-prone, and insufficiently scalable to meet industrial production demands. This paper proposes and evaluates an automated vision-based robotic sorting system designed to enforce quality control in compliance with the Philippine National Standards (PNS). The system integrates a six-degree-of-freedom (6DOF) robotic arm, an Arduino Mega 2560 microcontroller, and a computer vision module employing the YOLOv11 object detection algorithm. Bananas are classified across four freshness categories—ripe, unripe, overripe, and defective—enabling precise acceptance or rejection decisions prior to processing. Among the evaluated YOLOv11 variants, the medium-scale YOLOv11m model demonstrated optimal performance, achieving 99.8% precision, 99.4% recall, and a Mean Average Precision (mAP@0.5) of 99.4% after 157 training epochs. Physical sorting trials validated the seamless integration of the vision pipeline with the robotic actuation system, confirming accurate pick-and-place operations without mechanical damage to the fruit. The proposed framework significantly reduces dependence on manual inspection, mitigates human error, and establishes a scalable foundation for intelligent automation in agricultural food processing.

Keywords— Computer Vision, Robotic Arm, YOLOv11, Banana Sorting, Quality Control, Agricultural Automation.

I. INTRODUCTION

Quality control is a foundational requirement of modern food manufacturing, ensuring product consistency and regulatory compliance throughout every stage of the supply chain. In the Philippines, banana ketchup is a widely consumed condiment whose commercial production depends critically on the consistent quality of its primary ingredient—the Saba banana (*Musa 'Saba'*). Raw material intake must adhere to the PNS/BAFS 41:2015 Code of Practice for the Processing of Banana Ketchup and the PNS/BAFPS 64:2008 Fresh Fruit Specification, which collectively define the physical and maturity parameters acceptable for processing.

Non-compliant batches directly degrade final product quality, increase waste, and reduce operational efficiency.

Despite advances in manufacturing technology, the banana ketchup industry continues to rely predominantly on traditional manual sorting. Human inspectors visually assess individual bananas for ripeness, surface defects, and structural integrity—a process that is inherently subjective and susceptible to fatigue-induced errors. During peak harvest periods, the throughput demands of sorting lines frequently exceed the reliable capacity of human workers, creating bottlenecks that compromise both production schedules and quality consistency.

The convergence of robotics and artificial intelligence presents a compelling alternative to these operational limitations. Computer vision systems can evaluate produce objectively and at high speed, classifying agricultural items based on color, texture, morphology, and surface characteristics without fatigue-related degradation. Among contemporary object detection architectures, YOLOv11 offers a favorable balance of inference speed and classification accuracy, making it well suited for real-time industrial deployment. Complementing the vision component, a 6DOF robotic arm provides the mechanical dexterity and workspace flexibility necessary to handle delicate fruit with precision and without inducing mechanical damage.

This study addresses the research gap between laboratory-grade computer vision research and practical, field-deployable agricultural automation systems. The specific objectives are: (1) to develop and train a YOLOv11-based classification model capable of distinguishing ripe, unripe, overripe, and defective Saba bananas; (2) to integrate this vision system with a 6DOF robotic arm through forward and inverse kinematic modeling; and (3) to validate the complete sorting pipeline under physical trial conditions. By achieving these objectives, this work contributes a replicable framework that can reduce manual labor dependency, enforce

PNS quality standards, and support the broader adoption of smart automation in Philippine food processing industries.

II. METHODOLOGY

This chapter details the researchers' methodical approach to the study, encompassing its pre-design, design, development, and testing phases, all crucial to its successful completion.

A. System Architecture

The proposed automated sorting system is organized as a sequential data pipeline comprising four primary subsystems: image acquisition, computer vision classification, kinematic computation, and robotic actuation. Figure 1 illustrates the overall architecture and data flow.

A high-resolution smartphone camera mounted overhead captures continuous real-time images of the sorting workspace. Captured frames are transmitted via a USB Type-C interface to a host laptop equipped with an AMD Ryzen 9 processor and an NVIDIA GeForce RTX 4070 GPU (8 GB VRAM). The host computer executes the YOLOv11 inference engine to classify detected bananas and subsequently applies a monocular depth estimation algorithm to derive three-dimensional spatial coordinates from the two-dimensional image plane. Computed joint angle targets are transmitted over a serial communication interface to an Arduino Mega 2560 microcontroller. The microcontroller generates Pulse Width Modulation (PWM) signals that drive six MG996R servo motors, directly commanding each joint of the 6DOF robotic arm.

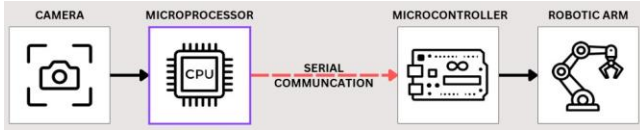


Figure 1: System architecture

B. Hardware and Mechanical Setup

The physical prototype is constructed on a custom plywood platform that supports both the 6DOF aluminum-frame robotic arm and an overhead camera mount. The operational workspace is constrained to one square foot (approximately 929 cm²), a dimension selected to ensure that the entire sorting area remains within the camera's field of view without parallax distortion. The camera is mounted parallel to the work surface to optimize the accuracy of two-dimensional coordinate extraction.

The sorting layout includes two functional zones: Container A serves as the intake basin holding unsorted mixed batches of Saba bananas, while Container B receives bananas that satisfy the quality criteria specified by PNS/BAFPS 64:2008. Bananas classified as overripe or defective are directed to alternative drop-off locations outside the primary acceptance zone. The adaptive gripper mechanism, attached to the arm's end-effector, is designed to accommodate the natural size variation of Saba bananas while exerting sufficient gripping force to achieve reliable pick-and-place operations without bruising the fruit.

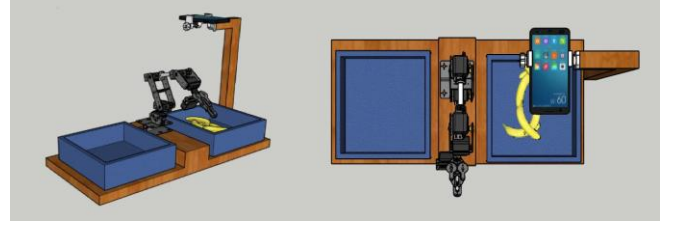


Figure 2: 3D Design

C. Software Implementation and Algorithms

The classification dataset comprises 120 annotated images of Saba bananas distributed equally across four categories: ripe, unripe, overripe, and rotten (defective). Images were annotated using Label Studio and subjected to data augmentation—including rotation, brightness adjustment, and horizontal flipping—to improve model generalizability and reduce overfitting. The augmented dataset was exported in YOLO format and used to train three model variants: YOLOv11n (nano), YOLOv11m (medium), and YOLOv11x (extra-large), all implemented within the PyTorch deep learning framework.

Spatial localization of detected objects is achieved through monocular depth estimation, which converts two-dimensional bounding box centroids returned by the YOLOv11 model into three-dimensional Cartesian coordinates relative to the robot's base frame. The Denavit-Hartenberg (D-H) convention was implemented in Python to define the kinematic chain of the 6DOF arm. Forward kinematics maps joint angle configurations to end-effector positions, while the inverse kinematic solver computes the joint angles required to position the gripper over a target object given its estimated spatial coordinates. This mathematical framework ensures positional accuracy during the pick-and-place cycle.

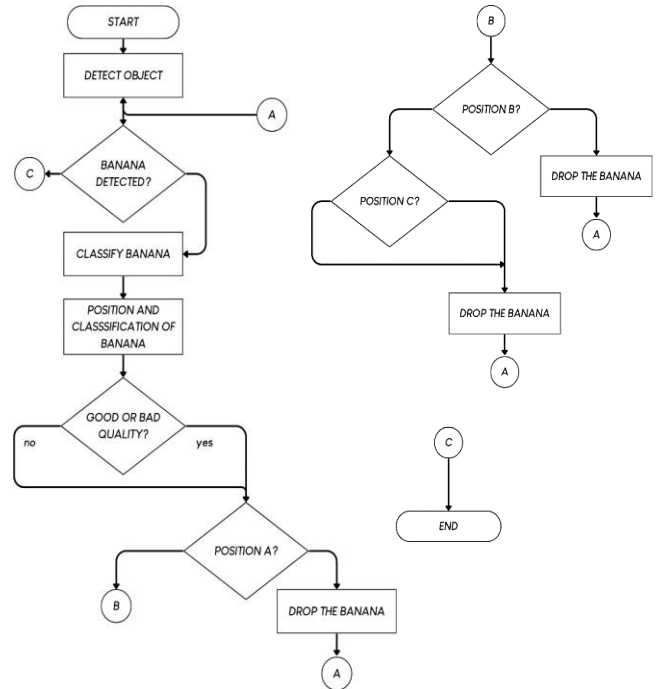


Figure 3: System Flowchart

D. Experimental Procedure

Each sorting trial begins with the robotic arm in a predefined rest position. Upon detection of a banana within the camera's field of view, the YOLOv11 model performs classification and returns a bounding box with an associated class label and confidence score. If the detected class satisfies the acceptance criteria (ripe), the kinematic solver computes the corresponding joint angles, and the arm executes a pick operation followed by placement at Drop-off Position A (Container B). Bananas classified as unripe, overripe, or defective are routed to designated alternative drop-off positions. The control loop iterates continuously until Container A is empty, at which point the arm returns to its rest position. Sorting accuracy was evaluated using the percentage of correct classifications (% CC) across all physical trials:

$$\% \text{ CC} = (\text{Number of Correct Classifications} / \text{Total Number of Samples}) \times 100\%$$

III. RESULTS & DISCUSSION

A. Comparative Evaluation of YOLOv11 Variants

Model selection required a principled trade-off among inference speed, classification accuracy, and GPU memory utilization. Three variants of YOLOv11 were systematically evaluated to identify the configuration that best satisfies the operational requirements of a real-time industrial sorting system.

The YOLOv11n (nano) variant delivered the highest inference speed among the three candidates but exhibited reduced discriminative capacity when distinguishing between visually similar subcategories within the acceptable class, particularly bananas transitioning between ripe and unripe states. This limitation reduced its reliability for strict quality gate applications.

The YOLOv11x (extra-large) variant achieved marginal improvements in recall over the medium model; however, its computational demand consistently exceeded the optimal VRAM allocation of the available NVIDIA GeForce RTX 4070 GPU, introducing inference latency that is incompatible with real-time sorting throughput requirements.

The YOLOv11m (medium) variant achieved the best balance of accuracy, speed, and resource efficiency. Trained over 157 epochs, it attained an overall precision of 99.8%, a recall of 99.4%, an F1-Score of 99.6%, and an mAP@0.5 of 99.4%. For the critical "Good" (ripe/acceptable) classification, the model recorded 99.9% precision and 99.3% mAP, demonstrating exceptional reliability for binary accept/reject decisions in quality control workflows. Model performance was evaluated using the following standard metrics:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$mAP = (1/n) \times \sum AP_k \text{ for } k = 1 \text{ to } n$$

Table 1 presents the complete comparative performance metrics for all three YOLOv11 variants across model-level and class-level evaluations.

TABLE I

Performance Evaluation Metrics Comparing YOLOv11n, YOLOv11m, and YOLOv11x Models

Model	Class	Precision	Recall	F1-Score	mAP @0.5
YOLOv11n	All	98.6%	97.8%	98.2%	98.1%
	Good	98.9%	97.5%	98.2%	98.0%
	Defective	98.3%	98.1%	98.2%	98.2%
YOLOv11m	All	99.8%	99.4%	99.6%	99.4%
	Good	99.9%	99.1%	99.5%	99.3%
	Defective	99.7%	99.7%	99.7%	99.5%
YOLOv11x	All	99.1%	98.7%	98.9%	98.8%
	Good	99.3%	98.5%	98.9%	98.7%
	Defective	98.9%	98.9%	98.9%	98.9%

B. System Integration and Physical Sorting Accuracy

The end-to-end integration of the YOLOv11 vision pipeline with the robotic actuation subsystem demonstrated high operational reliability across all physical sorting trials. Three-dimensional coordinate estimates derived from monocular depth estimation were accurately translated into joint angle commands via the D-H kinematic model, enabling the robotic arm to execute precise pick-and-place operations with consistent positional repeatability.

The adaptive gripper successfully handled Saba bananas across the natural range of sizes encountered in the test batches without inflicting bruising, surface compression, or other forms of mechanical damage—a critical requirement for maintaining the commercial integrity of accepted fruit prior to ketchup processing. This outcome validates the suitability of the MG996R servo-driven gripper design for delicate agricultural manipulation tasks.

Physical sorting trials confirmed that the system accurately identified and isolated bananas exhibiting a range

of quality defects, including windscars, bird damage, and premature or uneven ripening. Accepted bananas—those classified as ripe and structurally sound—were reliably transferred to the designated processing container (Container B), while non-compliant specimens were redirected to alternative drop-off zones without manual intervention.

The automated sorting pipeline represents a substantive operational improvement over conventional manual inspection. Human sorters are subject to fatigue-induced inconsistency, particularly during extended shifts or high-throughput periods, whereas the proposed system maintains constant classification performance independent of processing duration. Furthermore, objective algorithmic classification eliminates the inter-inspector variability inherent to human visual assessment, providing a reproducible and auditable quality gate that aligns directly with PNS/BAFPS 64:2008 compliance requirements.

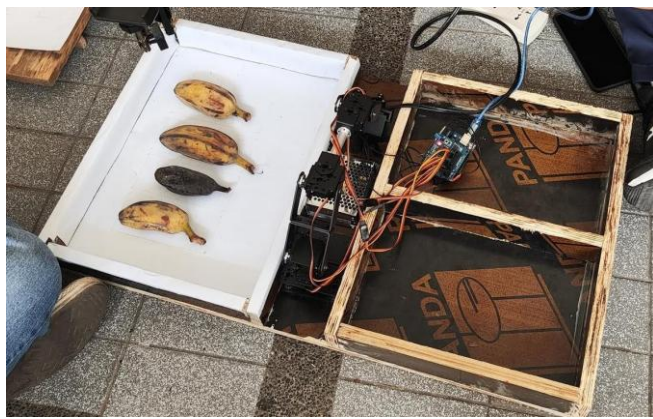


Figure 4: Actual Prototype Setup

IV. CONCLUSION

This study has successfully demonstrated the design, implementation, and experimental validation of a vision-based robotic sorting system tailored for the quality control of Saba bananas destined for ketchup manufacturing. The system integrates YOLOv11m-based computer vision, monocular depth estimation, Denavit-Hartenberg kinematic modeling, and a 6DOF servo-driven robotic arm into a cohesive and automated sorting pipeline.

The YOLOv11m model, selected through systematic comparative evaluation of three YOLOv11 variants, achieved outstanding classification performance: 99.8% precision, 99.4% recall, and a mAP@0.5 of 99.4% across all four banana freshness categories. Physical sorting trials confirmed the reliable translation of vision outputs into accurate robotic actions, with correct classification and placement of both acceptable and defective specimens. These results establish the proposed framework as a technically viable and practically deployable alternative to manual sorting in the banana ketchup manufacturing sector.

The system's primary contributions include: (1) the elimination of subjectivity and fatigue-related errors from the quality inspection process; (2) enforcement of PNS/BAFS 41:2015 and PNS/BAFPS 64:2008 compliance through consistent algorithmic classification; and (3) a replicable

integration architecture connecting deep learning inference with physical robotic manipulation.

Future work should prioritize three development directions. First, the training dataset should be expanded to encompass edge-case samples such as partially defective specimens or bananas with non-standard morphologies, improving model robustness under real-world variability. Second, the kinematic algorithms should be refined to increase movement fluidity and reduce per-cycle processing time, enabling higher throughput in large-scale industrial deployments. Third, upgrading to GPU hardware with greater VRAM capacity will allow evaluation of larger model architectures—such as YOLOv11x—that may offer further improvements in recall for atypical defect categories. Collectively, these advancements will accelerate the transition toward fully automated, standards-compliant smart processing lines in Philippine food manufacturing.

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